UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

END SEMESTER EXAMINATION
SEMESTER II - 2023/2024
PHY1203/PH 102 - General Physics II

Time allowed: ***Three** hours*

Answer **FIVE** questions selecting **TWO** from Section A, **TWO** from Section B and **ONE** from Section C.
(Use of calculators is allowed)
Symbols have their usual meanings unless specified otherwise

Electron charge (e) $= 1.60 \times 10^{-19}$ C
Mass of electron (m_e) $= 9.11 \times 10^{-31}$ kg
Mass of proton (m_p) $= 1.67 \times 10^{-27}$ kg
Avogadro number (N_A) $= 6.02 \times 10^{23}$ mol^{-1}
Permittivity of free space (ε_o) $= 8.85 \times 10^{-12}$ F m^{-1}
Permeability of free space (μ_o) $= 4\pi \times 10^{-7}$ H m^{-1}
Stefan-Boltzmann constant (σ) $= 5.67 \times 10^{-8}$ W m^{-2} K^{-4}

Planck's constant (h) $= 6.63 \times 10^{-34}$ J s
Speed of light (c) $= 3.00 \times 10^{8}$ m s^{-1}
Wien's constant $= 2.89 \times 10^{-3}$ m K

Section-A

01. (a) An electric dipole consists of two-point charges of equal magnitude and opposite polarity separated by a small distance, as shown in Figure 1.

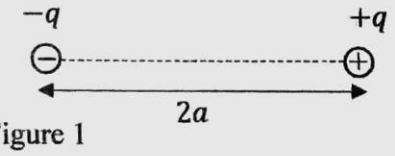


Figure 1

(i) Define the dipole moment of a dipole.

(ii) Write an expression for the dipole moment vector $\vec{p}$. Define all the symbols used and state the direction of $\vec{p}$.

(iii) A dipole is placed in a uniform external electric field $\vec{E}$, making an angle θ with the field as shown in Figure 2.

1. Derive an expression for the torque (τ) acting on the dipole.

2. Show that $\vec{\tau} = \vec{p} \times \vec{E}$.

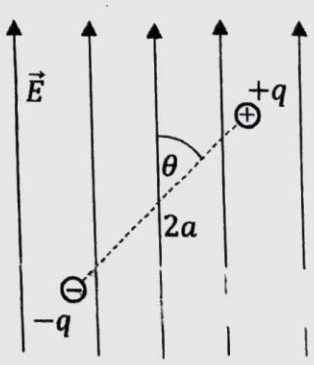


Figure 2

(b) When a dipole in an electric field changes its orientation, the field exerts a torque that performs work on it, resulting in a change in its potential energy. For an electric dipole in an external electric field $\vec{E}$, the potential energy is given by $U = -(\vec{p} \cdot \vec{E})$.

Consider an electric dipole consisting of charges $\pm 1.6 \times 10^{-19}$ C separated by 0.125 nm, placed in a uniform electric field of 5×10^5 N C^{-1}, as shown in the Figure 3.

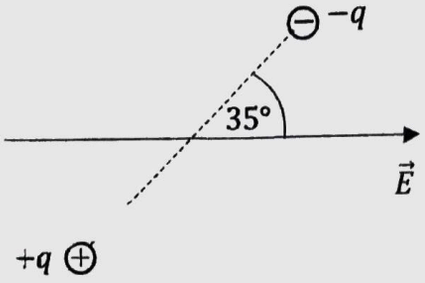


Figure 3

(i) Determine the net force exerted by the field on the dipole.

(ii) Find the magnitude and direction of the electric dipole moment.

(iii) Determine the magnitude and direction of the torque acting on the dipole.

(iv) Estimate the potential energy of the dipole in the position shown.

02. (a) (i) State Gass's Theorem.

(ii) A long straight cylindrical conductor with radius a is placed horizontally and has a linear charge density of $+\lambda$.

(1) Define linear charge density.

(2) Obtain an expression for the electric field intensity E at a point having a distance $r(r \geq a)$ from the central axis in terms λ, r and ε_0. Where ε_0 is the permittivity of vacuum.

(3) Draw a graph representing the variation of electric field intensity E with respect to the distance (r) from the ass of the conductor.

(b) An electron is placed at a point r distance from the axis of the charged conductor described in question (ii) above. Consider the charge and mass of the electron are e and m respectively.

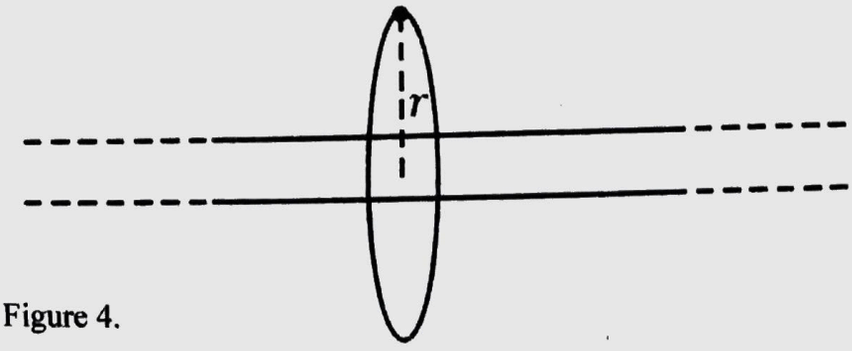


Figure 4.

If the electron moves in a circular path with a radius r centered on the axis of the conductor as shown in Figure 4, find the following (neglect the effects of gravity):

(i) Copy Figure 4 onto your answer script, indicate the force acting on the electron, and obtain the magnitude of the force in terms of λ, r, ε_0 and e

(ii) Obtain an expression for the angular velocity of the election in terms of λ, r, ε_0, e and m.

(iii) Write the expression for the tangential speed of electron in term of in terms of λ, r, ε_0, e and m.

(iv) Evaluate the period of the electron. Given that the linear charge density in SI units $\lambda = 6 \times 10^{-15}$ and distance from the axis $r = 5\ \mu$m.

03. (a) (i) State Ampère's circuital law and write it in integral form, clearly defining all symbols used.

(ii) Using Ampère's law, derive an expression for the magnetic field due to a long straight conductor carrying a steady current I.

(iii) A long straight wire carries a current of 5.0 A. Calculate the magnitude of the magnetic field at a point located 2.0 cm from the wire.

(b) (i) An infinitely long straight cylindrical conductor of radius a carries a steady current I along its axis. The current is distributed over the cross-section so that the current density J may be uniform or vary with the radial distance r from the axis.

(1) Assuming uniform current density J throughout the conductor, derive expressions for the magnetic field $B(r)$ inside $(r < a)$ and outside $(r > a)$ the conductor using Ampère's law.

(2) Sketch the variation of $B(r)$ with r.

(ii) Now suppose the current density varies with radius as $J(r) = J_0(1 - \frac{r}{a})$ where J_0 is the maximum current density at the center of the conductor.

(1) Find an expression for the total current I in the conductor.

(2) Using Ampère's law, derive expressions for the magnetic field $B(r)$ inside $(r < a)$ and outside $(r > a)$ the conductor.

(3) Show that your results reduce to those in part (b)(i)(1) when $J(r)$ is constant.

(iii) Discuss how a non-uniform current distribution affects the magnetic field inside the conductor compared to the uniform case.

Section-B

04. (a) Using the Stefan–Boltzmann law of blackbody radiation, write down an expression for the emissive power (P) of a blackbody at temperature T.

(b) The total emissive power of a blackbody is 1.29×10^{6} W m^{-2}. Find the temperature of the body.

(c) Write down Wien's displacement law as an equation and identify each symbol.

(d) Sketch the blackbody radiation curves for four different temperatures of the blackbody.

(e) If the surface temperature of a star is 10,000 K, calculate the wavelength corresponding to the peak intensity of the blackbody radiation.

(f) Identify the region of electromagnetic spectrum to which the above wavelength belongs.

(g) Determine the energy of a photon (in eV) corresponding to the above peak wavelength.

05. (a) (i) Write down Einstein's photoelectric equation and identify each symbol.

(ii) Light of wavelength 400 nm falls on a metallic surface and ejects electrons. If the work function of the metal is 2.0 eV calculate the followings.

(1) The threshold wavelength.
(2) The maximum kinetic energy of the emitted electrons.
(3) The stopping potential.

(b) With the aid of a clearly labelled diagram of an X-ray tube, explain how X-rays are produced.

(i) Sketch the X-ray spectrum and giving reasons, explain the shape of the spectrum.

(ii) Derive an expression for the minimum wavelength (λ_{min}) of X-rays in terms of applied tube potential difference V. Clearly identify the other symbols you use.

(iii) An X-ray tube operates at 50 kV. Calculate the minimum wavelength (in nm) of X-rays produced.

(iv) Explain how X-rays are used for crystal structure analysis.

06. (a) (i) Write down the radioactive decay law as an equation in terms of activity A.

(ii) Obtain the relationship between the decay constant (λ) and half-life ($T_{1/2}$) of a radioactive substance.

(iii) With the aid of nuclear reactions briefly explain the formation of carbon-14 in the atmosphere and how it is used to determine the age of an archaeological sample.

(iv) The activity of 1 g of carbon from an archaeological sample is 0.05 Bq, while the activity of 1 g of carbon from a present-day carbon sample is 0.12 Bq. Given that the half-life of carbon-14 is 5730 years, calculate the age of the archaeological sample.

(b) In a nuclear reactor the following nuclear reaction is taking place.

$$^{235}_{92}\mathrm{U} + \mathrm{n} \rightarrow {}^{141}_{a}\mathrm{X} + {}^{92}_{36}\mathrm{Kr} + b\mathrm{n} + \mathrm{Q}$$

Here a and b are integers. The atomic masses of the elements and the neutron are given bellow.

$^{235}_{92}\mathrm{U} = 235.043\ 93$ u $n = 1.008\ 67$ u

$^{141}_{a}\mathrm{X} = 140.914\ 41$ u $^{92}_{36}\mathrm{Kr} = 91.926\ 15$ u

Also 1 u = 931.5 MeV/c^2.

(i) What are the values of a and b?

(ii) Calculate the mean binding energy of $^{235}_{92}\mathrm{U}$.

(iii) Also calculate the value of Q factor.

(iv) If 1.00 kg of uranium-235 is consumed per year in a nuclear reactor, calculate the power output of the reactor.

Section-C

07. (a) (i) Name **three** primary bond types and **three** secondary bond types.

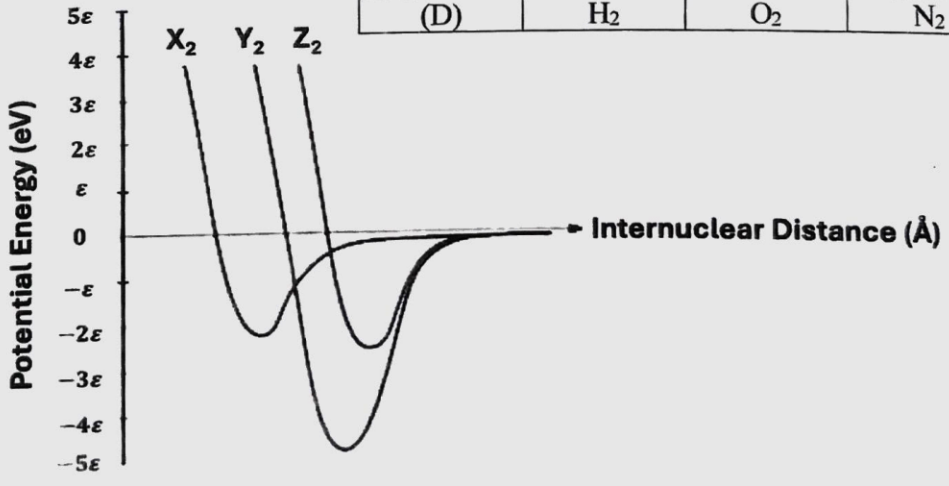
(ii) Briefly explain the difference between the primary bonds and the secondary bonds

(b) (i) State **three** physical properties of bulk materials that can be understood using the potential energy curve of a diatomic molecule.

(ii) Explain how you would utilize the features of the potential energy curve to understand each of the physical properties which you have stated above.

(iii) Potential energy as a function of internuclear distance for three diatomic molecules, X_2, Y_2, and Z_2 is shown in the figure below. Which of the following option correctly identifies the diatomic molecules, X_2, Y_2, and Z_2? Give reason for your selection.

	X_2	Y_2	Z_2
(A)	N_2	O_2	H_2
(B)	H_2	N_2	O_2
(C)	O_2	H_2	N_2
(D)	H_2	O_2	N_2



(c) The Lennard–Jones potential for two Nickel (Ni) atoms in a lattice can be written as follows:

$$V(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

where σ and ε are constants and r is the separation between the atoms.

(i) Identify the attractive and the repulsive potentials and comment on their origin.

(ii) Show that σ can be expressed as $\sigma = \left(\frac{1}{2}\right)^{\frac{1}{6}} r_0$, where r_0 is the separation of the atoms at the equilibrium.

(iii) For Ni, σ has a value of 1.5808 Å. Find the equilibrium separation of the Ni atoms in the lattice.

(iv) Derive an expression for the effective spring constant, k of Ni atoms in the lattice.

08. Assume that the bond length of $^{35}Cl_2$ molecule is 1.98 Å and the atomic mass of the ^{35}Cl isotope is 34.968853 u [1 u = 1.66×10^{-27} kg].

(a) (i) Calculate the energy of the first excited rotational state ($J = 1$) and the second excited rotational state ($J = 2$) of $^{35}Cl_2$ molecule

(ii) Find the wavelength of the radiation emitted in the transition from ($J = 2$) to ($J = 1$).

(iii) Consider the diatomic molecule of ^{37}Cl isotope ($^{37}Cl_2$ molecule) having the same bond length as $^{35}Cl_2$. Qualitatively compare the radiation emitted in the transition of $^{37}Cl_2$ from ($J = 2$) to ($J = 1$) with that of $^{35}Cl_2$, calculated in (ii).

(b) (i) Write down an expression for the energy of the nth vibrational states (or level) of a diatomic molecule and obtain an expression for the energy difference between two consecutive energy states. Define all symbols used.

(ii) The effective spring constant of $^{35}Cl_2$ molecule is 323 N m^{-1}. Assuming that the vibrational motion of this molecule obeys the classical simple harmonic motion, find the amplitude of oscillation when the molecule is in its first excited vibrational state ($v = 1$).

(c) The effective spring constant of (I_2) is 169 N/m. Molecular weight of I_2 is 126.905u. When compared to $^{35}Cl_2$ molecule, which molecule will absorb higher energy photons when transit between consecutive vibrational energy levels? Justify your answer.

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